

PropertyIQ Analytics Studio

Predictive Analytics System Performance Report

1. Executive Summary

PropertyIQ Analytics Studio is an end-to-end predictive analytics system designed to analyze real-estate property data, identify market patterns, generate data-driven insights and support property valuation using machine learning.

The analytical pipeline includes data cleaning, exploratory data analysis, visualization, predictive model training, model evaluation and automated report generation.

2. Dataset Overview

Total cleaned property records: 199

Total analytical features: 5

Average property price: \$428,062.07

Median property price: \$416,776.00

Average property area: 2,220 square feet

3. Data Cleaning and Preprocessing

Remaining null values after preprocessing: 0

Remaining duplicate records: 0

The preprocessing pipeline handles missing values, duplicate records and numerical outliers before the dataset is used for visualization and machine learning.

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4. Exploratory Data Analysis

Exploratory analysis was performed to understand property price distributions, relationships between property characteristics and market segmentation patterns.

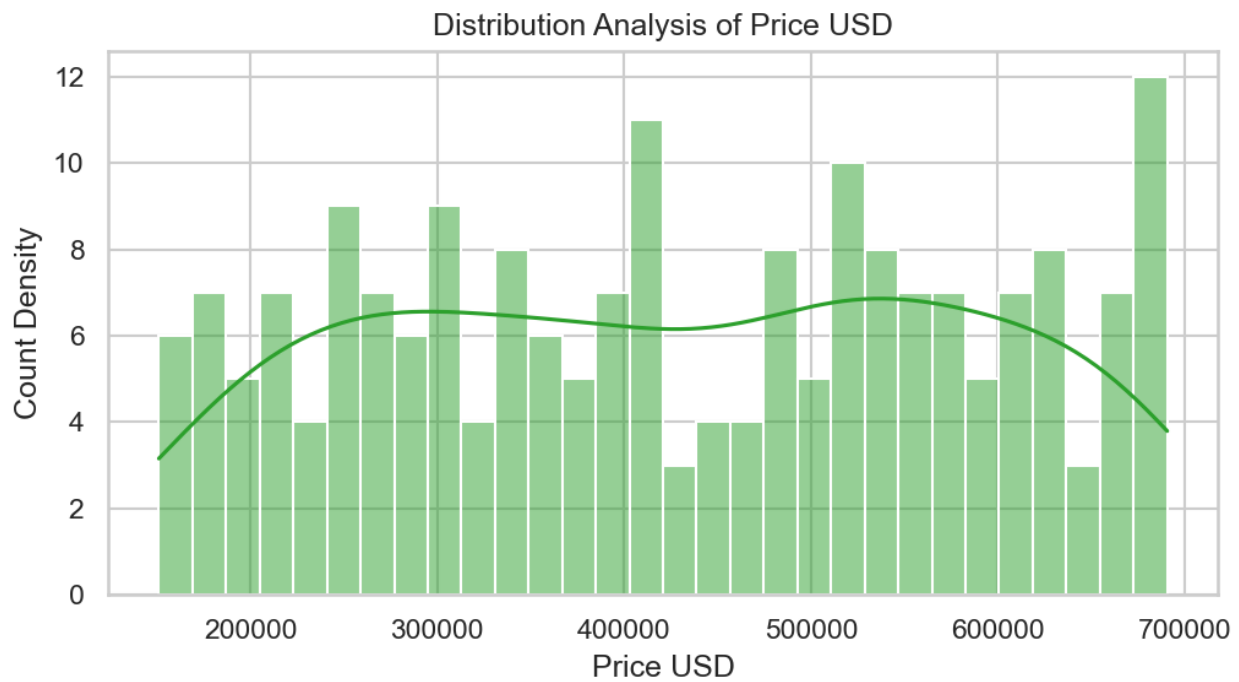
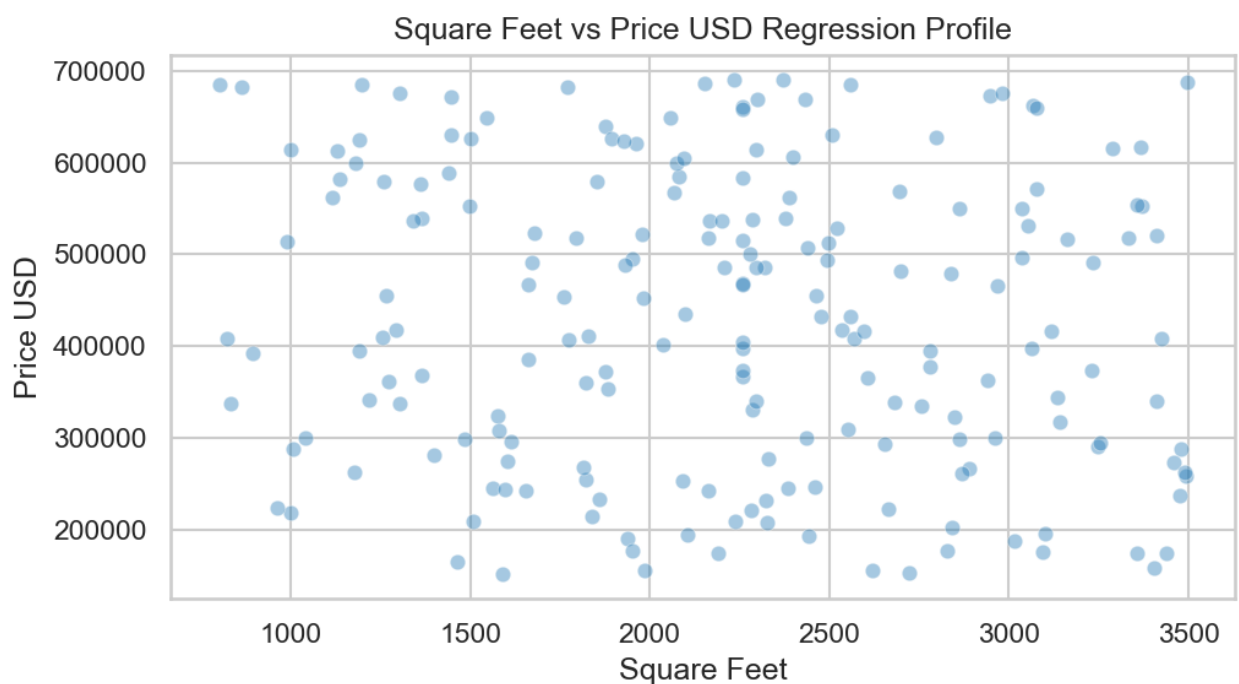


Figure 1: Property price distribution analysis.



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Figure 2: Relationship between property area and market price.

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5. Predictive Modeling

The predictive analytics pipeline uses regression-based machine learning techniques to estimate property prices from available property characteristics.

Linear Regression is used as a baseline model, while Random Forest Regression provides a more advanced non-linear modeling approach.

The dataset is divided into training and testing subsets. Model performance is evaluated using metrics such as R-squared and Mean Absolute Error.

6. Business Insights

Location Grade C currently has the highest average property value at approximately \$449,288.31.

Property valuation should be interpreted using multiple features together, including property area, bedroom configuration, property age and location characteristics.

7. Conclusion

PropertyIQ Analytics Studio demonstrates a complete predictive analytics workflow, beginning with raw data processing and progressing through visualization, machine learning and actionable insight generation.

The interactive Streamlit dashboard extends the analytical pipeline by allowing users to explore data, generate property predictions, compare scenarios and review model intelligence through a unified interface.